# King County
# House Price Analysis

*Predicting real estate prices in King County, WA*

May 2014 – May 2015 · 21,613 transactions · 21 features

| 21,596 | 24.6% | $450K | $R^2$ 0.74 |
|---|---|---|---|
| Clean Records | Luxury (≥$650K) | Median Price | Best Model |

# Agenda



**01**

## Data Overview & Cleaning

21,613 records · targeted outlier removal · 0 nulls

**02**

## EDA & Key Findings

Distributions, correlations, density lens, geo map

**03**

## Feature Engineering

7 new features: house_age, renovation flags, ratios

**04**

## Preprocessing

Log-transform, StandardScaler, 80/20 split

**05**

## Model Comparison

5 models: Linear ⟶ Gradient Boosting

**06**

## Results & Luxury Segment

$R^2$=0.74 overall · $R^2$=0.56 luxury · feature importance

# 01 · Data Overview & Cleaning



## Dataset Facts

| | |
|---|---|
| **Records** | 21,613 (21,596 after cleaning) |
| **Features** | 21 raw → 19 used in model |
| **Period** | May 2014 – May 2015 |
| **Missing values** | No imputation needed |
| **Price min** | $78,000 |
| **Price median** | $450,000 |
| **Price mean** | $540,198 |
| **Price max** | $7,700,000 |

## Cleaning Decisions

### Removed

- bedrooms = 33 — data entry error (sqft_living doesn't match)
- bedrooms = 0 or bathrooms = 0 — likely land plots, not houses

### Kept as-is

- sqft_basement = 0 — normal, many houses have no basement
- yr_renovated = 0 — normal, means never renovated
- Large lots, high prices — real market data, not errors

*Targeted cleaning based on domain logic — not blanket IQR removal.*

# 02 · EDA — Key Findings



### 📈 Price is Right-Skewed

**Skewness: 4.026**

Log-transform reduces to 0.430 — needed for linear models

### 🏠 sqft_living Dominates

**r = 0.702**

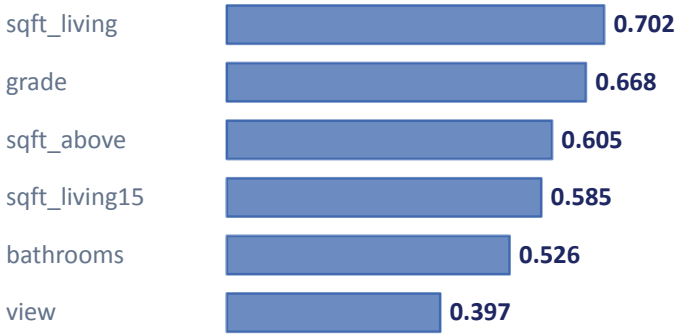
Highest linear correlation with price among all raw features

### 🌊 Waterfront Premium

**~3× price**

Waterfront properties command dramatically higher prices

## Top Correlations with Price

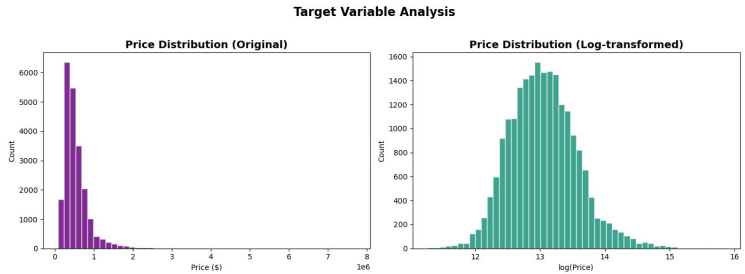


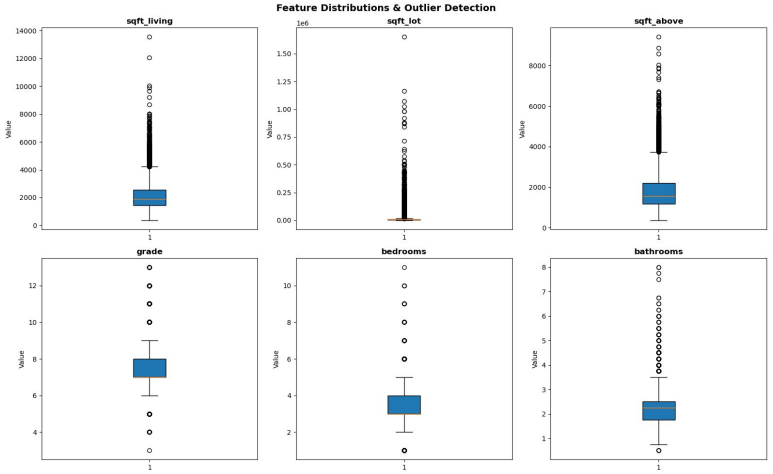

## Luxury vs Standard Segment

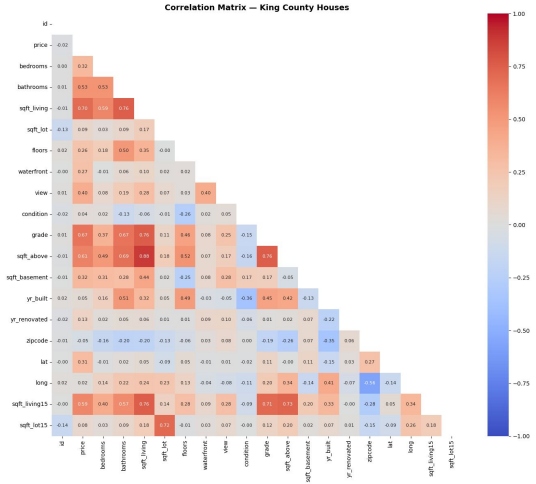
| | | |
|---|---|---|
| **Count** | 16,274 (75.4%) | 5,322 (24.6%) |
| **sqft_living avg** | 1,680 sqft | 2,895 sqft |
| **Grade avg** | 7.1 | 9.2 |
| **Bathrooms avg** | 1.9 | 2.8 |
| **Waterfront** | 0.4% | 4.1% |

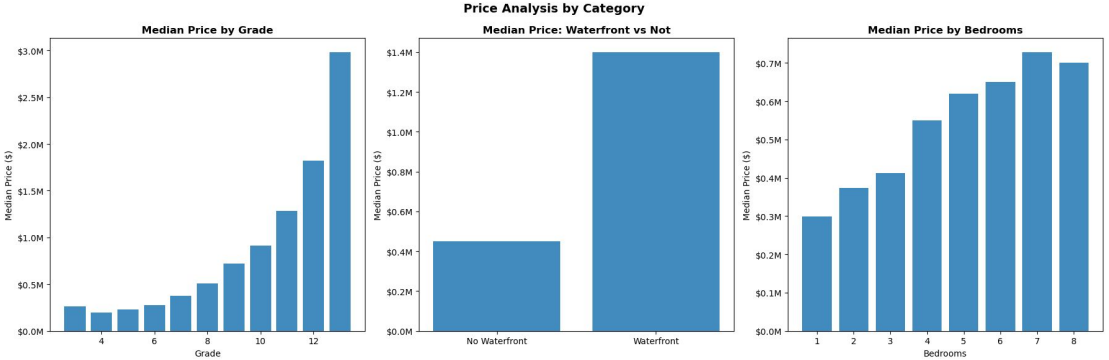
# 02. EDA - Visualizations





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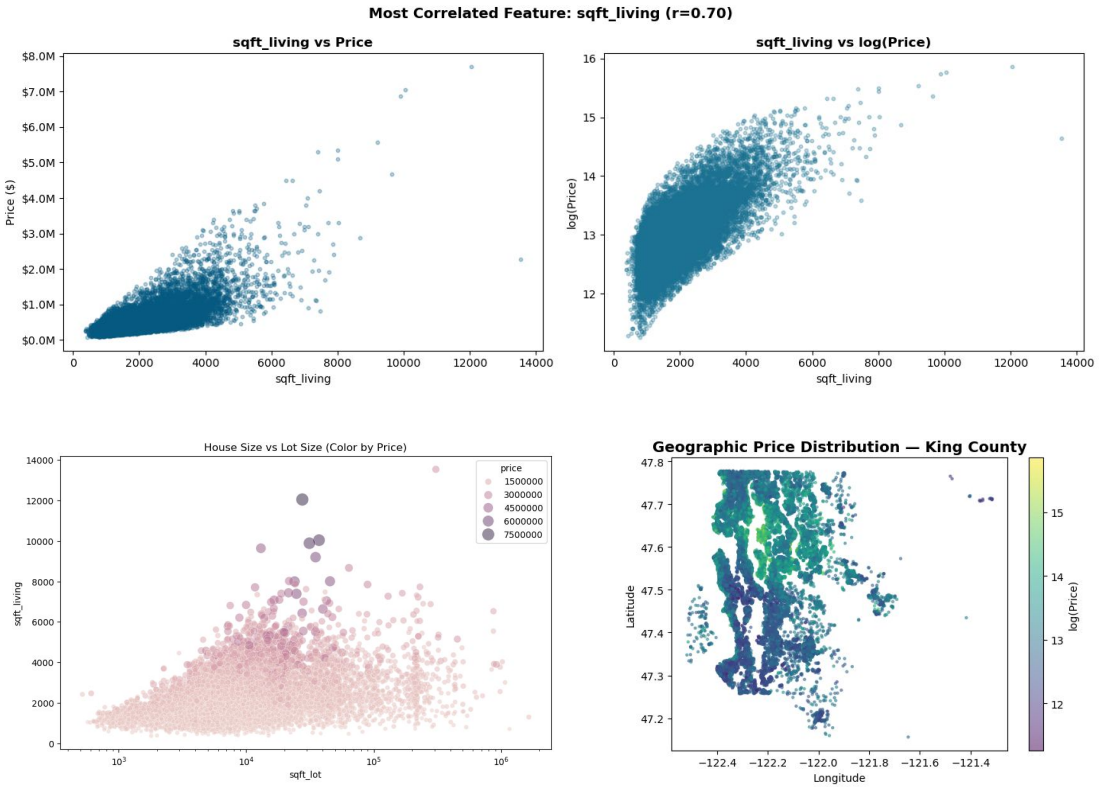


# 03–04 · Feature Engineering & Preprocessing



## 7 New Features Created

| | |
|---|---|
| `house_age` | sale_year − yr_built |
| `was_renovated` | Binary: renovated or not |
| `years_since_reno` | Recency of last renovation |
| `has_basement` | Binary: basement present |
| `living_vs_neighbors` | sqft_living / sqft_living15 |
| `sale_month` | Seasonal price variation |
| `lot_utilization` | *EDA only — not used in model* |

*price_per_sqft also computed for EDA only — both excluded to avoid data leakage.*

## Preprocessing Pipeline

**1 — Log-transform target**

Skewness: 4.026 → 0.430
Back-transform with np.expm1() for metrics

**2 — Train / Test Split**

80% train (17,276) · 20% test (4,320)
random_state=42 for reproducibility

**3 — StandardScaler**

Fit on train ONLY → transform both
Prevents data leakage from test set

**4 — No null imputation**

0 missing values in dataset
Step skipped entirely

# 05 · Model Selection & Comparison



*5 models trained on log(price) — metrics evaluated on back-transformed dollar predictions*

| | $R^2$ Score (higher = better) | RMSE & MAE |
|---|---|---|
| **Gradient Boosting** | **0.7408** | RMSE: $184,198 \| MAE: $119,308 |
| Random Forest | 0.7268 | RMSE: $189,096 \| MAE: $114,498 |
| Lasso | 0.7048 | RMSE: $196,553 \| MAE: $129,807 |
| Ridge | 0.7037 | RMSE: $196,930 \| MAE: $129,942 |
| Linear Regression | 0.7037 | RMSE: $196,941 \| MAE: $129,944 |

**Why Gradient Boosting?** Sequentially corrects errors from prior trees. Captures non-linear interactions (e.g. grade × sqft). No distribution assumptions. +14pp $R^2$ vs linear baseline.

# 06 · Results, Feature Importance & Luxury Segment



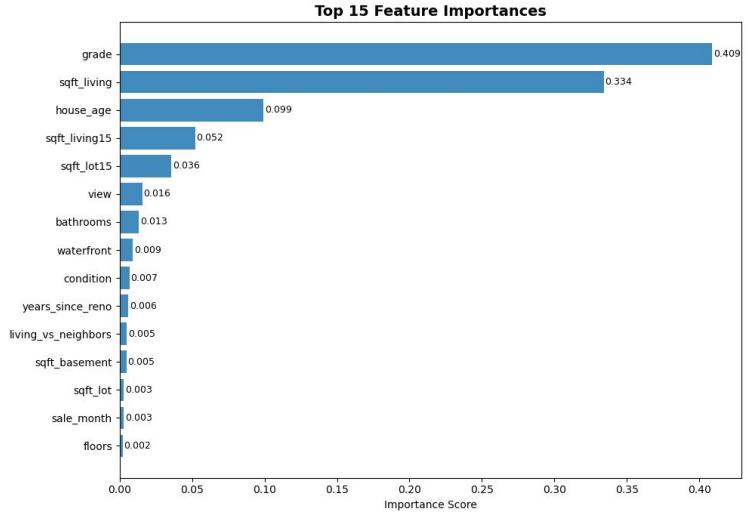
**0.7408**
**$R^2$ Overall**
*74% variance explained*

**$184K**
**RMSE Overall**
*On original price scale*

**0.5631**
**$R^2$ Luxury ≥$650K**
*1,025 test houses*

**$314K**
**RMSE Luxury**
*Higher variance segment*

## Feature Importances (Gradient Boosting)




## Key Insights

⭐ grade ranks #1 in importance (40.9%) despite r=0.668 — non-linear multiplier effect

📐 sqft_living is #2 (33.4%) — consistent with highest linear correlation (r=0.702)

🏗️ house_age (#3, 9.9%) not in raw data — confirms value of feature engineering

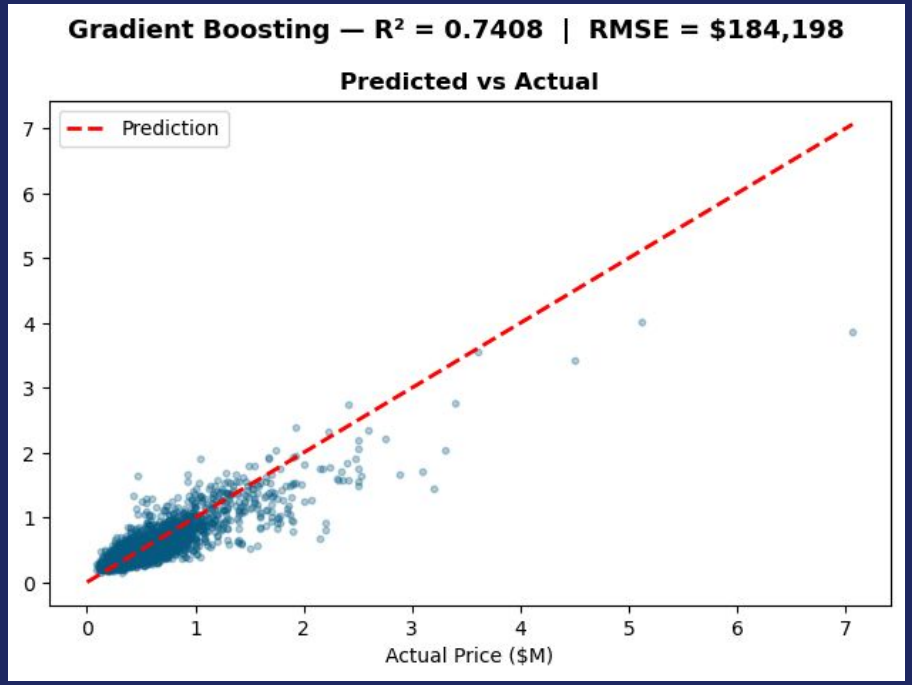
🌊 waterfront only 0.9% importance despite 3× price premium — rare in test set (n<50)

⚠️ Luxury $R^2$ drops from 0.74 → 0.56 — idiosyncratic pricing not captured by standard features

💡 Separate dedicated model recommended for the luxury segment

# Conclusions

- ✓ Gradient Boosting best model: $R^2$=0.74, RMSE=$184K
- ✓ grade is the #1 driver (40.9%) — non-linear multiplier effect
- ✓ Log-transforming price is critical: skewness 4.026 → 0.430
- ✓ house_age (#3) validates feature engineering value
- ✓ Luxury segment (24.6%) harder to predict: $R^2$=0.56
- ✓ Targeted cleaning beats blanket outlier removal




*Dataset: Kaggle — King County Houses · minasameh55*